

# AS-Level Maths:

## Core 2

*for Edexcel*



## C2.8 Integration



These icons indicate that teacher's notes or useful web addresses are available in the Notes Page.

This icon indicates the slide contains activities created in Flash. These activities are not editable.

For more detailed instructions, see the *Getting Started* presentation.

- **Evaluating definite integrals**
- The area under a curve
- The trapezium rule
- Examination-style questions

# Definite and indefinite integrals

When we find the **indefinite integral** of a function the answer takes the form of an expression plus a constant of integration.

When we find a **definite integral** of a function we find the integral between two given limits to give a numerical result.

For example, the definite integral of  $6x^2 + 5$  between  $x = 1$  and  $x = 3$  is written as

$$\int_1^3 (6x^2 + 5)dx$$

It is evaluated as follows:

$$\begin{aligned}\int_1^3 (6x^2 + 5)dx &= \left[ 2x^3 + 5x + c \right]_1^3 && \text{Notice that square brackets are used here ...} \\ &= (2(3)^3 + 5(3) + c) - (2(1)^3 + 5(1) + c) && \text{...and rounded brackets here.} \\ &= 69 - 7 && c \text{ cancels out.} \\ &= 62\end{aligned}$$

When evaluating a definite integral we can ignore constants of integration because they cancel out.

In general:

$$\int_a^b f'(x)dx = [f(x)]_a^b = f(b) - f(a)$$

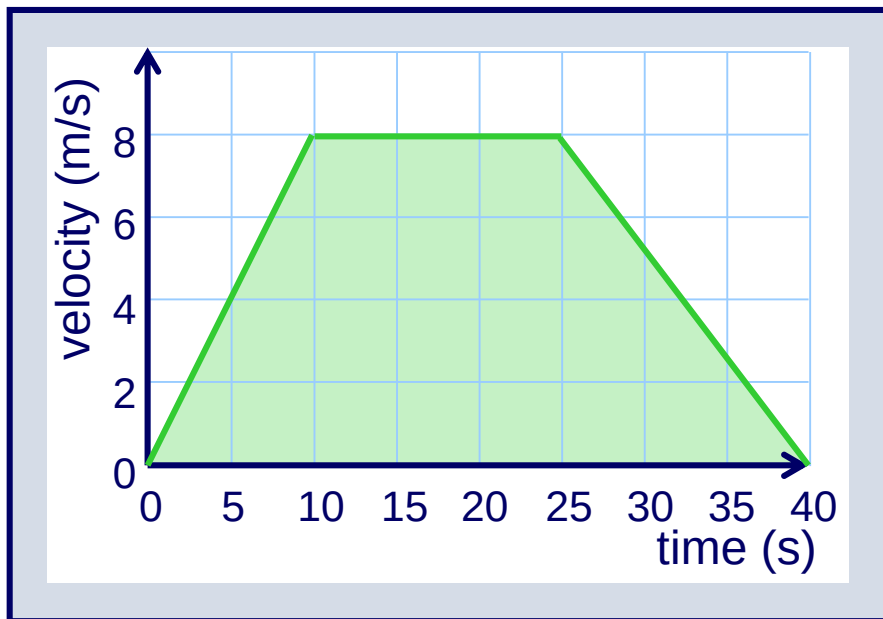
Evaluate  $\int_1^3 x^2 dx$

$$\begin{aligned}\int_1^3 x^2 dx &= \left[ \frac{x^3}{3} \right]_1^3 \\ &= \frac{(3)^3}{3} - \frac{(1)^3}{3} \\ &= 9 - \frac{1}{3} \\ &= 8\frac{2}{3}\end{aligned}$$

- Evaluating definite integrals
- **The area under a curve**
- The trapezium rule
- Examination-style questions

# The area under a velocity-time graph

The following graph shows how the velocity of a car changes over a period of 40 seconds.



What does the area under the graph represent?

Consider the area of a thin strip of the graph.

If the strip is thin enough it can be treated as if it is rectangular.

The area of the strip = velocity  $\times$  time = distance

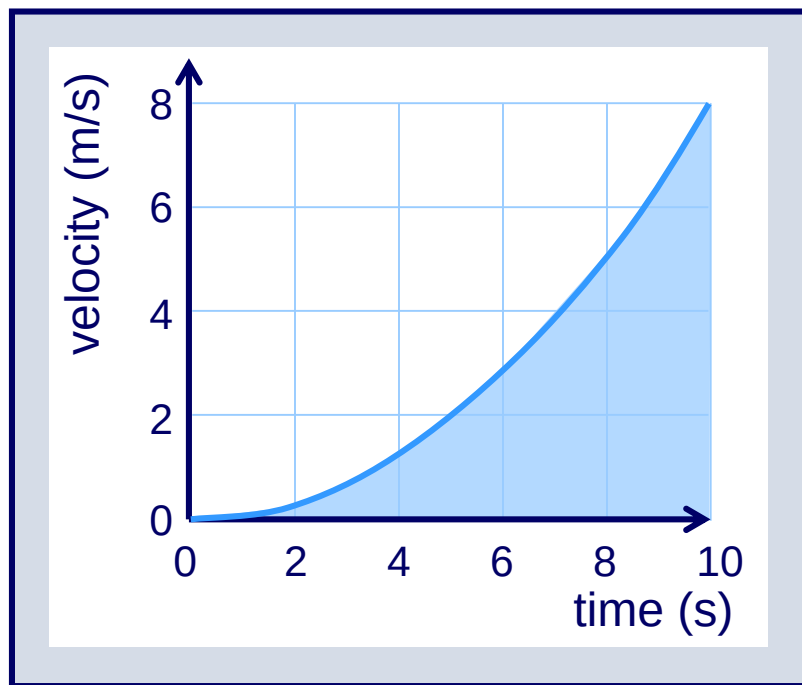
The area of each strip represents the distance travelled during a small period of time.

$\therefore$  The total area is the total distance travelled.

# The area under a velocity-time graph

In many situations the acceleration will not be constant and the velocity-time graph will be curved.

For example, this graph shows the velocity-time graph for a car in the first 10 seconds after it moves off from rest.



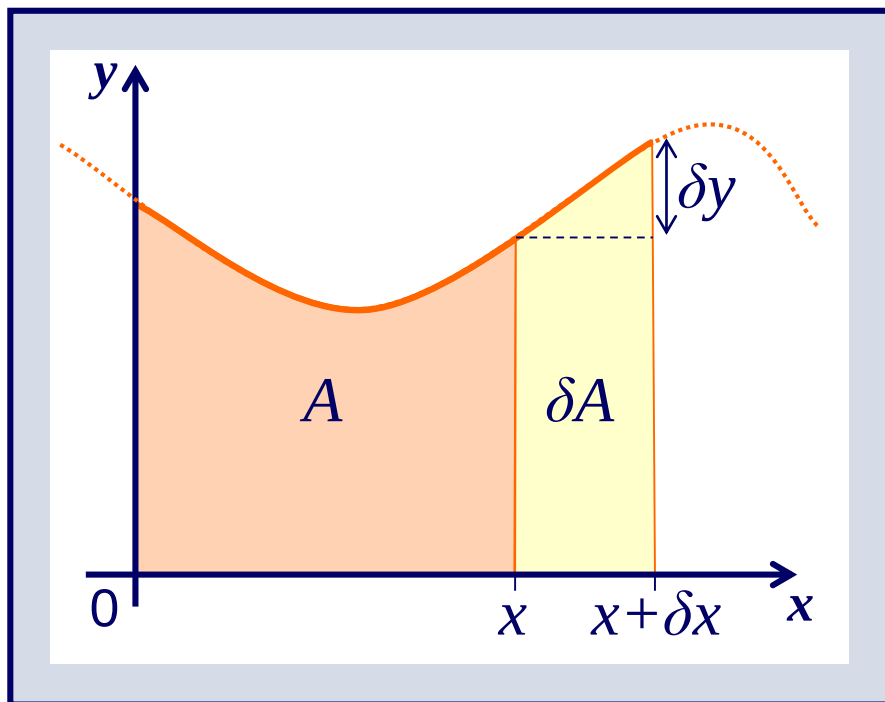
To find the distance travelled by the car over this time we need to find the area under the curve.

It is possible to estimate this area by dividing it into thin vertical strips and summing their areas.

However, if we know the velocity of the car as a function of time we can find the area more accurately by integration.

# The area under a curve

To see why the area under a curve can be found by integration, consider a continuous curve  $y = f(x)$ .



Let  $A$  be the area shown between the curve  $y = f(x)$  and the  $x$ -axis from  $x = 0$  to  $x$ .

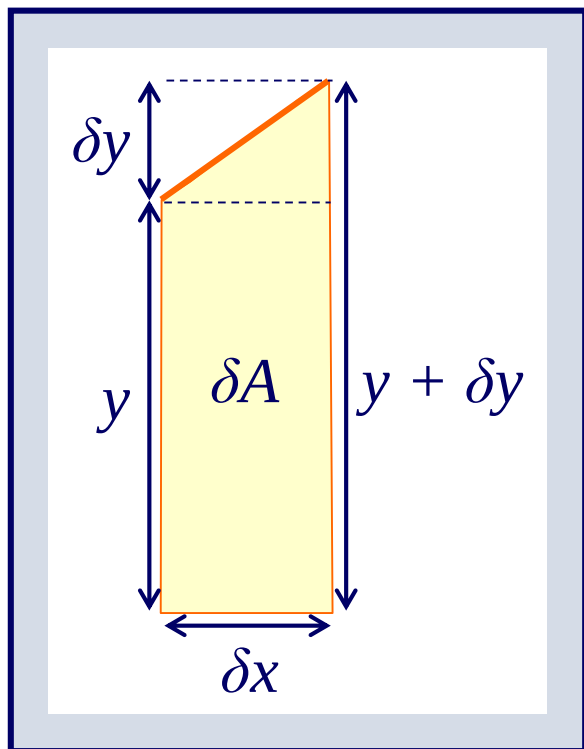
Now, imagine moving a small distance along the  $x$ -axis,  $\delta x$ .

We can call the corresponding increase in area  $\delta A$  and corresponding increase in  $y$   $\delta y$ .

Let's look more closely at this increase in area.

# The area under a curve

We can see that  $\delta A$  must be between  $y\delta x$  and  $(y + \delta y)\delta x$ .



We can write this as

$$y\delta x < \delta A < (y + \delta y)\delta x$$

Dividing through by  $\delta x$  gives

$$y < \frac{\delta A}{\delta x} < y + \delta y$$

Now consider the limit as  $\delta x$  and  $\delta y \rightarrow 0$ :

$$\lim_{\delta x \rightarrow 0} \frac{\delta A}{\delta x} = y \quad \text{so} \quad \frac{dA}{dx} = y$$

Integrating both sides with respect to  $x$  gives

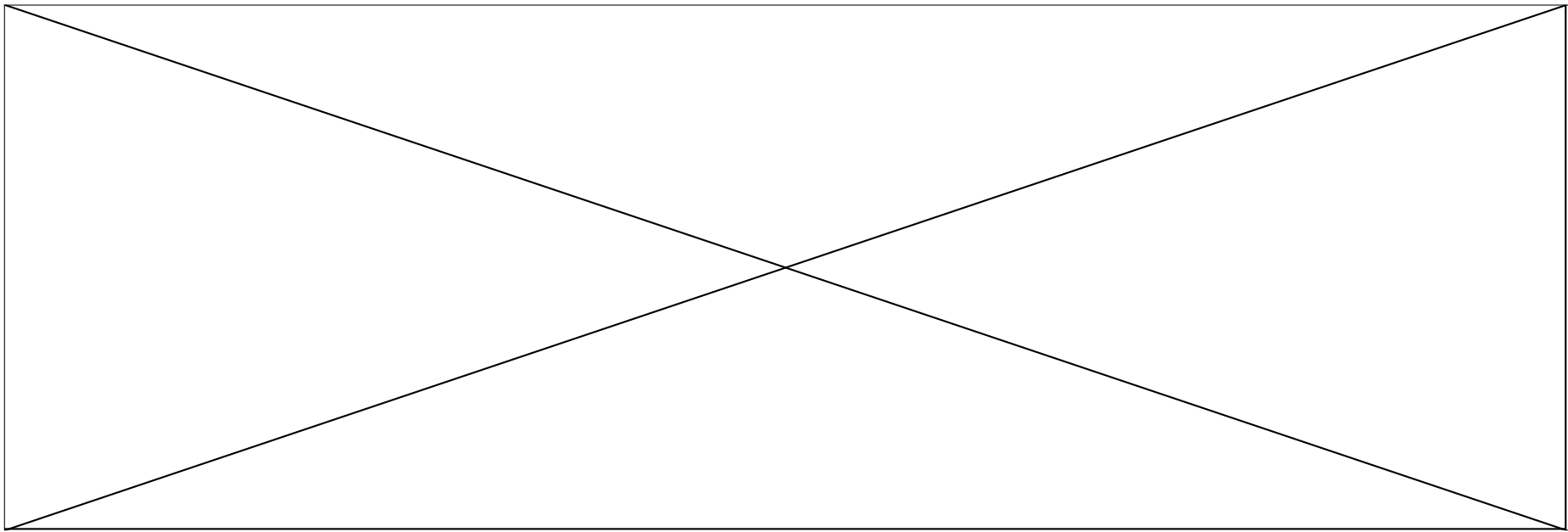
$$A = \int y dx$$

# The area function

The area under a curve  $y = f(x)$  is therefore a function of  $x$ , which we can write as  $A(x)$ .

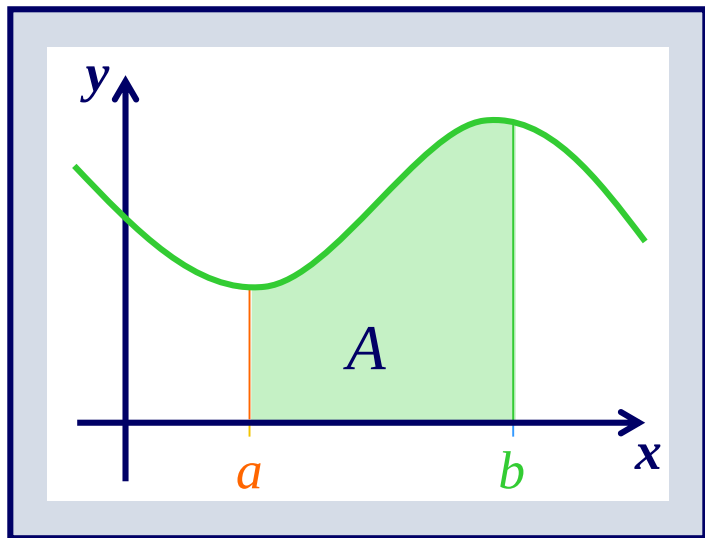
$A(x)$  is the area under the curve from  $x = 0$  to  $x$ .

Look what happens when the area function  $y = A(x)$  is plotted for some function  $y = f(x)$  as  $x$  varies.



# The area under a curve

Suppose that we want to find the area  $A$  under a curve between two given values of  $x$ ;  $x = a$  and  $x = b$ .



If the area between  $x = 0$  and  $b$  is  $A(b)$  and the area between  $x = 0$  and  $a$  is  $A(a)$

then the area  $A$  under the curve from  $x = a$  to  $x = b$  must be

$$A = A(b) - A(a)$$

In general:

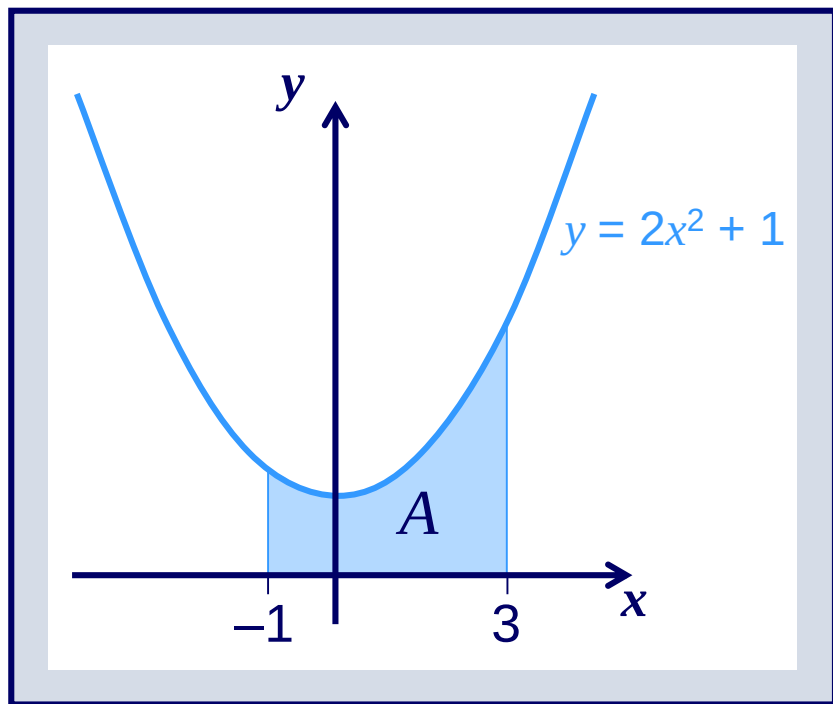
The area  $A$  under a curve  $y = f(x)$  between the limits  $x = a$  and  $x = b$  is given by

$$A = \int_a^b y \, dx$$

# The area under a curve

Find the area bounded by the curve  $y = 2x^2 + 1$ , the  $x$ -axis and the lines  $x = -1$  and  $x = 3$ .

Start by sketching the graph of  $y = 2x^2 + 1$  and shading the required area.

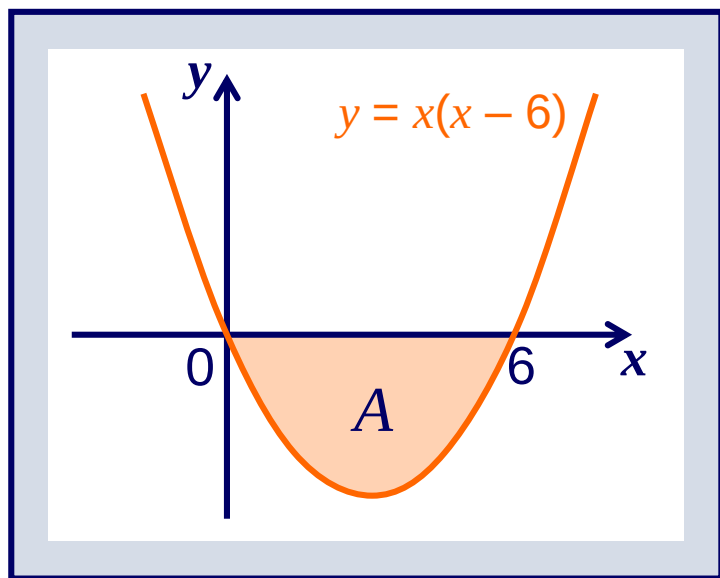


$$\begin{aligned} A &= \int_{-1}^3 (2x^2 + 1) dx \\ &= \left[ \frac{2x^3}{3} + x \right]_{-1}^3 \\ &= \left( \frac{2(3)^3}{3} + 3 \right) - \left( \frac{2(-1)^3}{3} - 1 \right) \\ &= 21 - -\frac{5}{3} \\ &= 22\frac{2}{3} \end{aligned}$$

The final solution can be written as  $22\frac{2}{3}$  units<sup>2</sup>.

Suppose the area that we are trying to find is *below* the  $x$ -axis.  
For example:

Find the area bounded by the curve  $y = x(x - 6)$  and the  $x$ -axis.



This area is given by:

$$\begin{aligned} A &= \int_0^6 (x^2 - 6x) dx \\ &= \left[ \frac{1}{3} x^3 - 3x^2 \right]_0^6 \\ &= \left( \frac{1}{3} (6)^3 - 3(6)^2 \right) - 0 \\ &= -36 \end{aligned}$$

The calculated area is negative because it is below the  $x$ -axis.

We write the actual area as 36 units<sup>2</sup>.

# Areas below the $x$ -axis

In some cases the area may fall partly below the  $x$ -axis and partly above. For example:

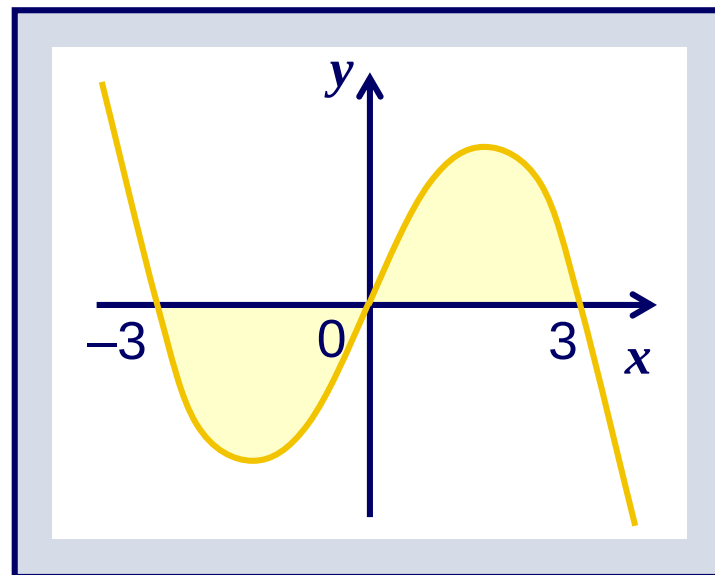
Find the area between  
 $y = 9x - x^3$  and the  $x$ -axis.

Factorizing gives

$$y = x(3 + x)(3 - x)$$

So the graph cuts the  $x$ -axis at  
 $x = -3$ ,  $x = 0$  and  $x = 3$ .

Also,  $x^3$  is negative so the curve is  $\cup$ -shaped.



The required area can now be shaded in.

Look what happens if we try to find the area under  $y = 9x - x^3$  between  $-3$  and  $3$ .

$$\begin{aligned}\int_{-3}^3 (9x - x^3) dx &= \left[ \frac{9}{2} x^2 - \frac{1}{4} x^4 \right]_{-3}^3 \\ &= \left( \frac{9}{2} (3)^2 - \frac{1}{4} (3)^4 \right) - \left( \frac{9}{2} (-3)^2 - \frac{1}{4} (-3)^4 \right) \\ &= 20\frac{1}{4} - 20\frac{1}{4} \\ &= 0\end{aligned}$$

From the shading on our graph we know the area isn't 0.

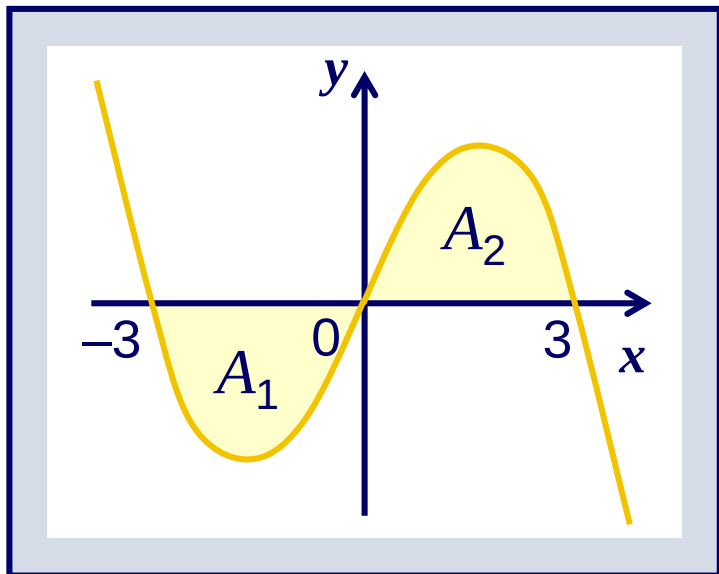
What has happened is that the negative area below the graph has cancelled out the positive area above the graph.

To avoid this we have to find the areas above and below the x-axis separately and then add the two areas together.



# Areas below the x-axis

Let's call the two areas  $A_1$  and  $A_2$ .



$$\begin{aligned} A_1 &= \int_{-3}^0 (9x - x^3) dx \\ &= \left[ \frac{9}{2} x^2 - \frac{1}{4} x^4 \right]_{-3}^0 \\ &= 0 - \left( \frac{9}{2} (-3)^2 - \frac{1}{4} (-3)^4 \right) \\ &= -20 \frac{1}{4} \end{aligned}$$

$$\begin{aligned} A_2 &= \int_0^3 (9x - x^3) dx \\ &= \left[ \frac{9}{2} x^2 - \frac{1}{4} x^4 \right]_0^3 \\ &= \left( \frac{9}{2} (3)^2 - \frac{1}{4} (3)^4 \right) - 0 \\ &= 20 \frac{1}{4} \end{aligned}$$

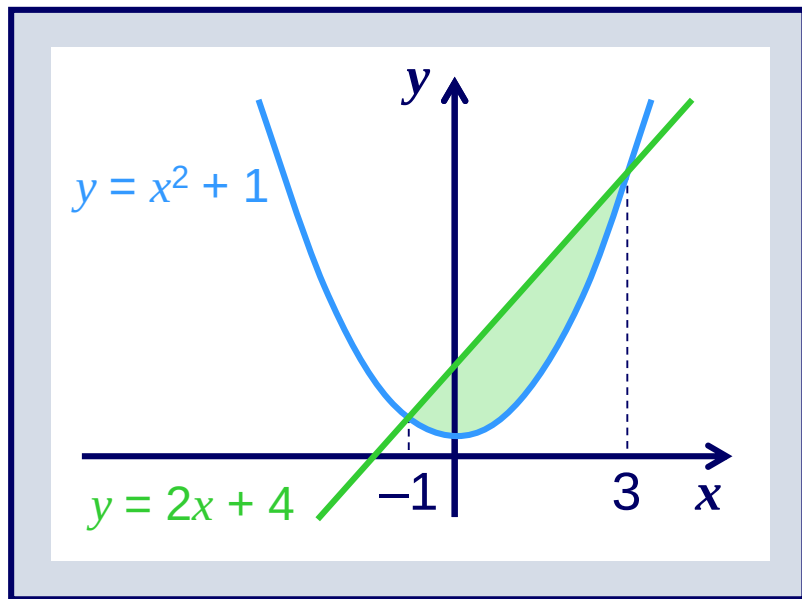
The total area =  $A_1 + A_2 = 40 \frac{1}{2}$  units<sup>2</sup>.

# The area between a curve and a line

Suppose we want to find the area between a curve and a line. For example:

Find the area bounded by the curve  
 $y = x^2 + 1$  and the line  $y = 2x + 4$ .

Start by sketching the graph, and shading the required area.



Next, we have to find the x-coordinates of the points where the curve and the line intersect.

$$\text{Solving: } x^2 + 1 = 2x + 4$$

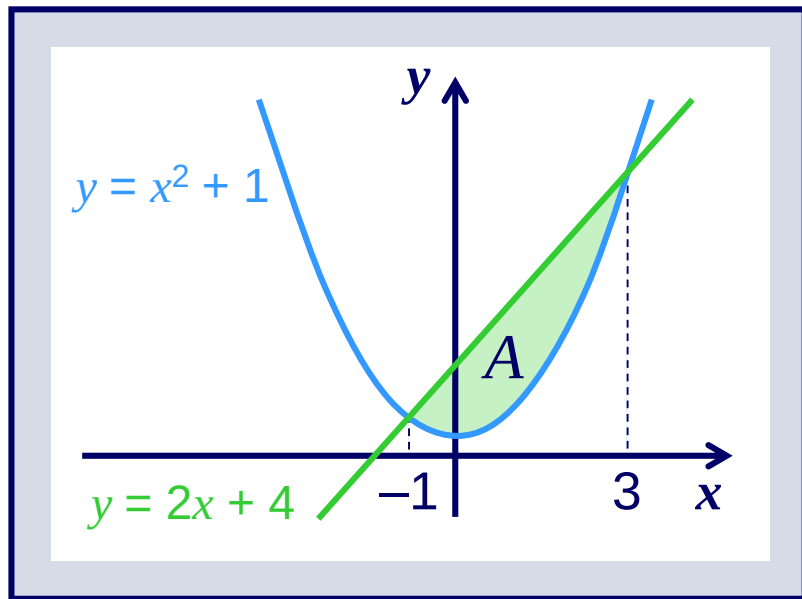
$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

  $\therefore$  the curve and the line intersect when  $x = -1$  and  $x = 3$ . 

# The area between a curve and a line

The area under the line  $y = 2x + 4$  between  $x = -1$  and  $x = 3$  is this area:



While the area under the curve  $y = x^2 + 1$  between  $x = -1$  and  $x = 3$  is this area:

So the area we require,  $A$ , can be found by subtracting the area under the curve from the area under the line.

This area is given by:

$$A = \int_{-1}^3 (2x + 4) dx - \int_{-1}^3 (x^2 + 1) dx$$

# The area between a curve and a line

Since the limits are the same these two integrals can be combined to give

$$\begin{aligned} A &= \int_{-1}^3 \left( (2x + 4) - (x^2 + 1) \right) dx \\ &= \int_{-1}^3 (-x^2 + 2x + 3) dx \\ &= \left[ -\frac{1}{3}x^3 + x^2 + 3x \right]_{-1}^3 \\ &= (-9 + 9 + 9) - \left( \frac{1}{3} + 1 - 3 \right) \\ &= 9 - 1\frac{2}{3} \\ &= 7\frac{1}{3} \end{aligned}$$

So the required area is  $7\frac{1}{3}$  units<sup>2</sup>.

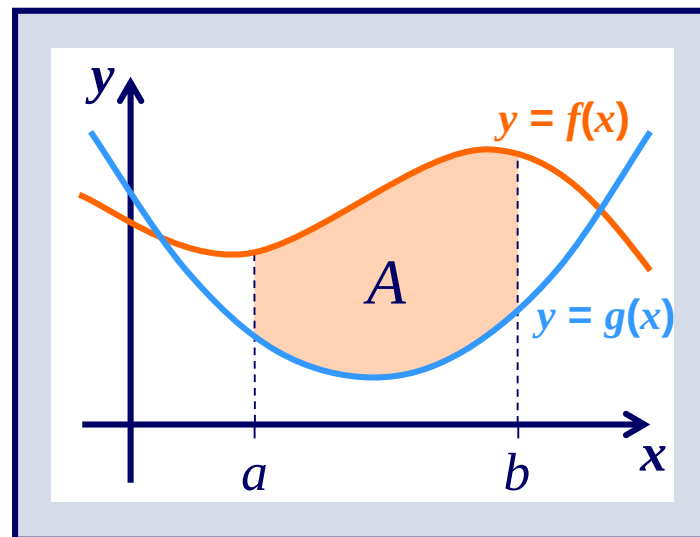
# The area between two curves

The method used to find the area between a curve and a line can also be used to find the area between two curves.

In general:

The area  $A$  between two curves  $y = f(x)$  and  $y = g(x)$  is given by

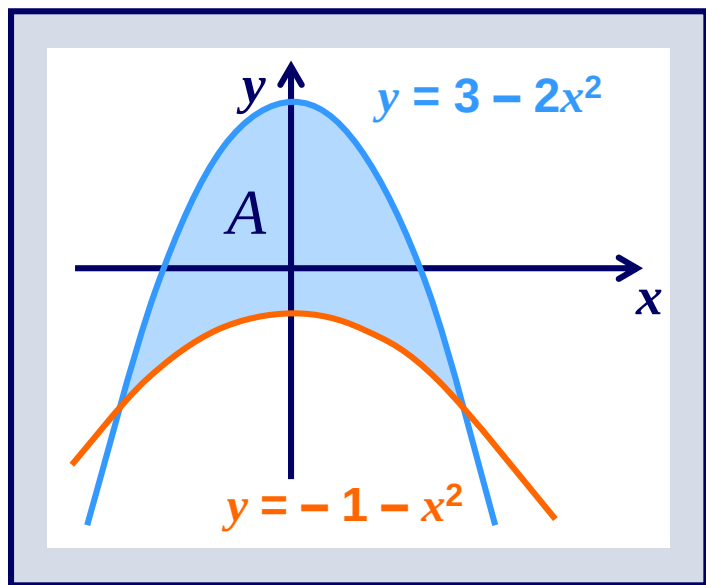
$$A = \int_a^b (f(x) - g(x)) dx$$



For example:

Find the area bound by the curves  
 $y = 3 - 2x^2$  and  $x = -1 - x^2$ .

# The area between two curves



Start by equating the curves to find out where they intersect.

$$3 - 2x^2 = -1 - x^2$$

$$4 - x^2 = 0$$

$$(2 - x)(2 + x) = 0$$

So the curves intersect at  $x = 2$  and  $x = -2$ .

Simplify the integral:

$$\begin{aligned} & \int_{-2}^2 \left( (3 - 2x^2) - (-1 - x^2) \right) dx \\ &= \int_{-2}^2 (4 - x^2) dx \end{aligned}$$

Find the area:

$$\begin{aligned} A &= \left[ 4x - \frac{1}{3}x^3 \right]_{-2}^2 \\ &= \left( 8 - \frac{8}{3} \right) - \left( -8 - \frac{-8}{3} \right) \\ &= 5\frac{1}{3} - -5\frac{1}{3} \\ &= 10\frac{2}{3} \text{ units}^2 \end{aligned}$$

- Evaluating definite integrals
- The area under a curve
- **The trapezium rule**
- Examination-style questions



# Approximating the area under a curve

Sometimes the area under a curve cannot be found by integration.

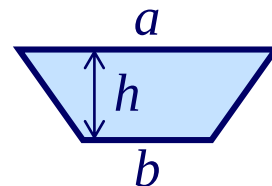
This may be because we cannot find the integral of the equation of the curve or because we need to find the area under a curve produced from experimental data.

In these cases we can use a method to approximate the area under the curve.

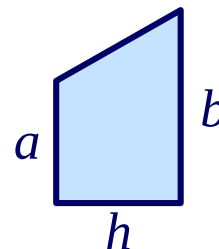
One such method is called **the trapezium rule**.

It works by dividing the area under a curve into trapeziums and calculating their areas. Remember:

$$\text{The area of a trapezium} = \frac{1}{2}(a + b)h$$

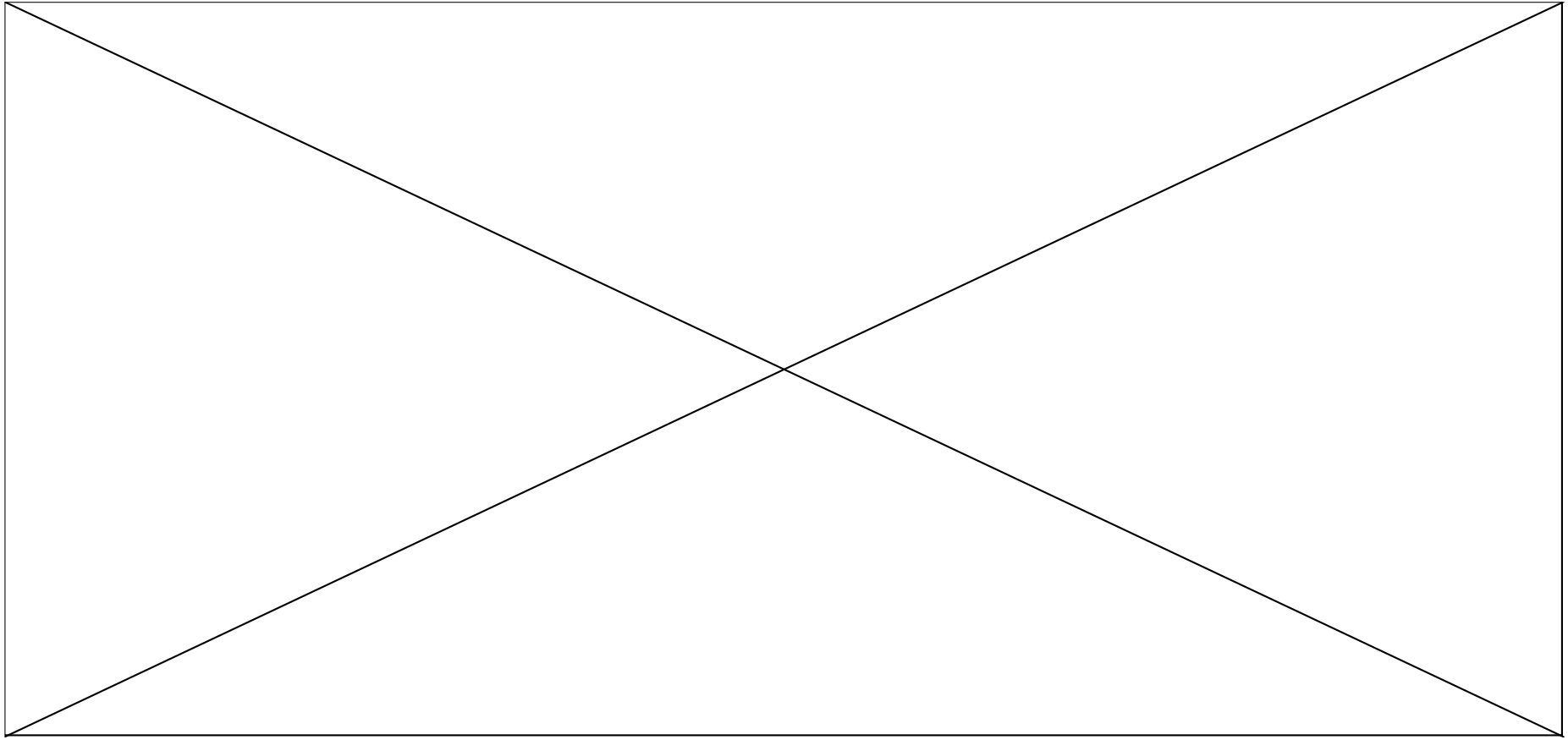


or



# The trapezium rule

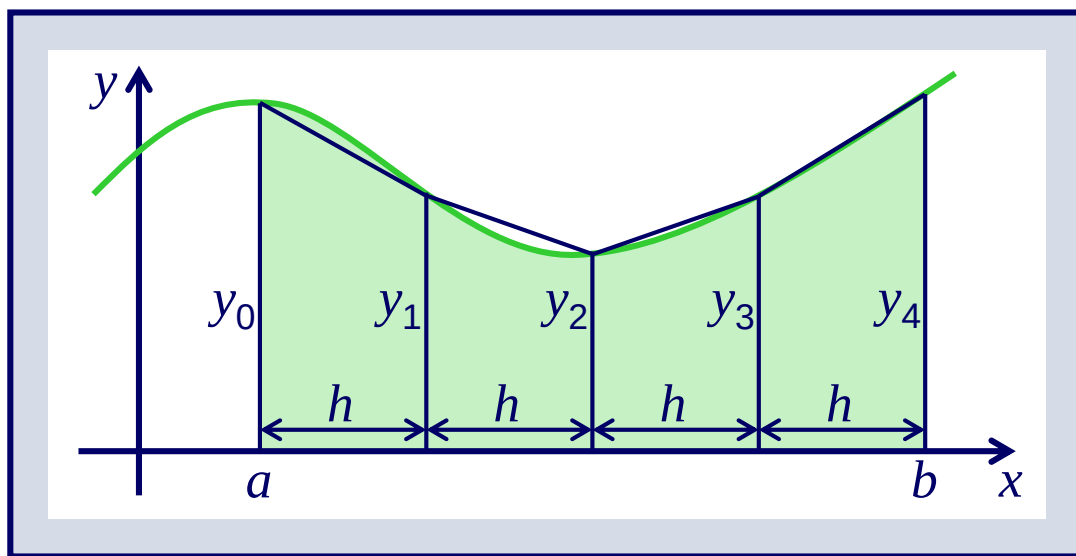
To demonstrate the method consider the area under the curve  
 $y = x^2 + 1$ .



# The trapezium rule

The more trapeziums the area is divided into the more accurate the estimate.

Suppose we wish to find the area under a curve  $y = f(x)$  between  $x = a$  and  $x = b$ .



We can divide the area into four trapeziums of equal width  $h$ .

The parallel sides of the four trapeziums are given by the five **ordinates**  $y_0, y_1, y_2, y_3$  and  $y_4$ .

In general, if there are  $n$  trapeziums there will be  $n + 1$  ordinates.

# The trapezium rule

The approximate area using the trapezium rule is:

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{1}{2}h(y_0 + y_1) + \frac{1}{2}h(y_1 + y_2) + \frac{1}{2}h(y_2 + y_3) + \frac{1}{2}h(y_3 + y_4) \\ &= \frac{1}{2}h(y_0 + y_1 + y_1 + y_2 + y_2 + y_3 + y_3 + y_4) \\ &= \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4)\end{aligned}$$

The ordinates have to be spaced out evenly so that the width of the trapeziums are the same.

$\therefore$  for  $n$  trapeziums of equal width  $h$ :  $h = \frac{b-a}{n}$

In general, the trapezium rule with  $n$  trapeziums is:

$$\int_a^b f(x)dx \approx \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$$

# The trapezium rule

Use the trapezium rule with four trapeziums to estimate the value of

$$\int_1^2 \frac{1}{x} dx$$

State whether this is an overestimate or an underestimate of the actual area.

The width  $h$  of each trapezium =  $\frac{2-1}{4} = 0.25$

We can use a table to record the value of each ordinate.

| $x$               | 1 | 1.25 | 1.5   | 1.75  | 2   |
|-------------------|---|------|-------|-------|-----|
| $y = \frac{1}{x}$ | 1 | 0.8  | 0.667 | 0.571 | 0.5 |

  $y_0$         $y_1$         $y_2$         $y_3$         $y_4$

# The trapezium rule

We can now work out the area using

$$\int_a^b f(x)dx \approx \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$$

with  $h = 0.25$  and the ordinates given by the table, so:

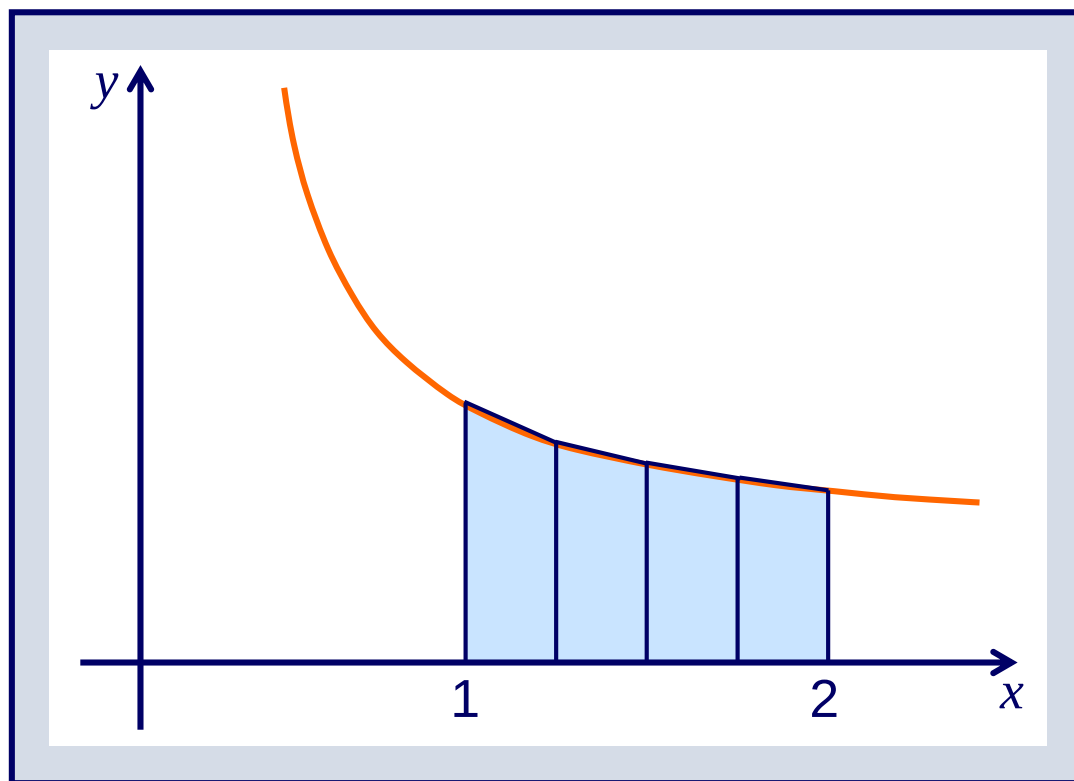
$$\begin{aligned}\int_1^2 \frac{1}{x} dx &\approx \frac{1}{2} \times 0.25 \times (1 + 2(0.8) + 2(0.667) + 2(0.571) + 0.5) \\ &= 0.70 \text{ (to 2 d.p.)}\end{aligned}$$

We can show whether this is an underestimate or an overestimate by sketching the area given by the trapezium rule on the graph of

$$y = \frac{1}{x}$$



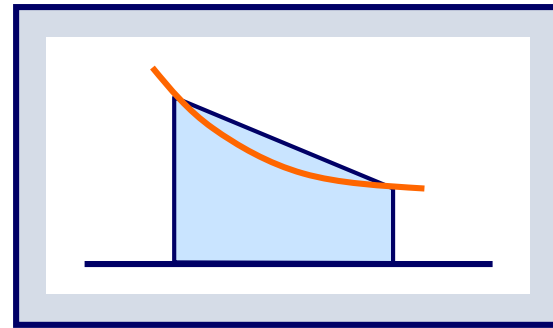
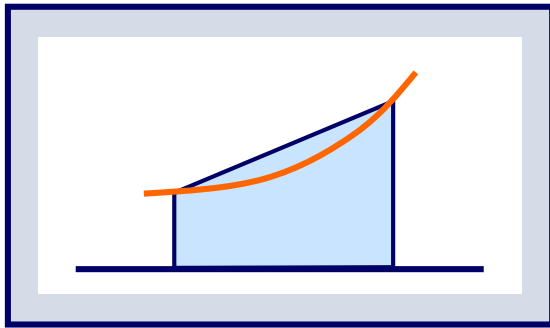
# The trapezium rule



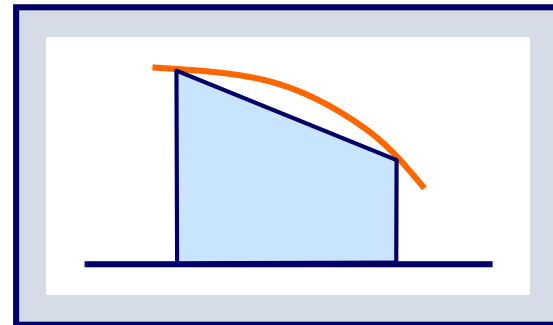
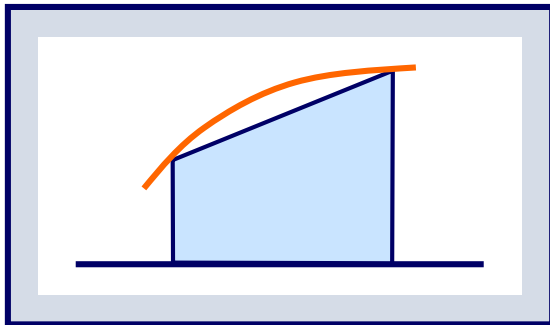
We can see from this sketch that the approximation given by the trapezium rule is a slight **overestimate** of the actual area.

# Overestimates and underestimates

In general, when the gradient of the graph is **increasing** over the given interval the area given by the trapezium rule will be an **overestimate** of the actual area.



When the gradient of the graph is **decreasing** over the given interval the area given by the trapezium rule will be an **underestimate** of the actual area.



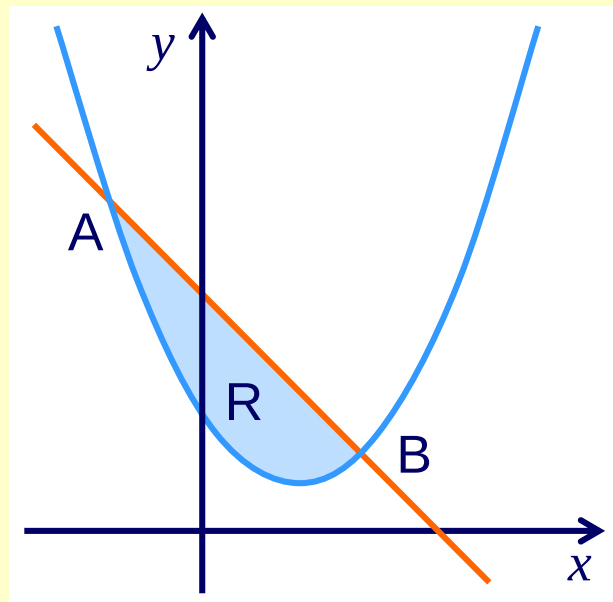
- Evaluating definite integrals
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# Examination-style question 1

The following diagram shows the curve with equation  $y = x^2 - 4x + 5$  and the line  $y = 8 - 2x$ .

The line and the curve intersect at the points A and B.



- Find the coordinates of A and B.
- Find the area of the shaded region R.

# Examination-style question 1

a) Equating the given equations:

$$x^2 - 4x + 5 = 8 - 2x$$

$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

$$x = -1 \quad \text{or} \quad x = 3$$

When  $x = -1$ ,  $y = 10$  and when  $x = 3$ ,  $y = 2$ .

$\therefore$  The coordinates of A are  $(-1, 10)$  and the coordinates of B are  $(3, 2)$ .

b) The area of R is given by

$$A = \int_{-1}^3 (8 - 2x)dx - \int_{-1}^3 (x^2 - 4x + 5)dx$$

# Examination-style question 1

Combining these gives

$$\begin{aligned} A &= \int_{-1}^3 (-x^2 + 2x + 3) dx \\ &= \left[ -\frac{1}{3}x^3 + x^2 + 3x \right]_{-1}^3 \\ &= (-9 + 9 + 9) - \left( \frac{1}{3} + 1 - 3 \right) \\ &= 9 - 1\frac{2}{3} \\ &= 7\frac{1}{3} \end{aligned}$$

So the required area is  $7\frac{1}{3}$  units<sup>2</sup>.



## Examination-style question 2

- a) Use the trapezium rule with 4 ordinates to estimate to 2 decimal places the value of

$$\int_0^{\frac{\pi}{3}} \tan x \, dx$$

- b) State whether the estimate given in part a) is an overestimate or underestimate of the area under the curve  $y = \tan x$  between  $x = 0$  and  $x = \frac{\pi}{3}$ .

The width  $h$  of each trapezium =  $\frac{\frac{\pi}{3}}{3} = \frac{\pi}{9}$

Using a table to record the value of each ordinate:

| $x$          | 0 | $\frac{\pi}{9}$ | $\frac{2\pi}{9}$ | $\frac{\pi}{3}$ |
|--------------|---|-----------------|------------------|-----------------|
| $y = \tan x$ | 0 | 0.364           | 0.839            | 1.732           |

# Examination-style question 2

We can now work out the area using

$$\int_a^b f(x)dx \approx \frac{1}{2}h(y_0 + 2y_1 + 2y_2 + \dots + 2y_{n-1} + y_n)$$

with  $h = \frac{\pi}{9}$  and the ordinates given by the table, so:

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \tan x dx &\approx \frac{\pi}{18} \times (0 + 2(0.364) + 2(0.839) + 1.732) \\ &= 0.72 \text{ (to 2 d.p.)}\end{aligned}$$

- b) Sketching the curve  $y = \tan x$  shows that the value given in a) is a slight overestimate of the actual area.

